

ANANYO DASGUPTA

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PROFILE SUMMARY

AI/ML student with hands-on experience in NLP, Neural Networks, Image Processing, and Computer Vision. Published researcher and multi-time hackathon winner, committed to building impactful, scalable solutions through innovation, collaboration, and learning.

SKILLS

Languages: Python, Java, R, SQL, C, MATLAB, HTML, CSS, JavaScript, TypeScript

Tools: Git, GitHub, Docker, Postman, Pinecone, MLFlow, Claude Code, GitHub Copilot, Firebase, MongoDB, Hugging Face

Frameworks: PyTorch, TensorFlow, Keras, OpenCV, Pandas, Matplotlib, Seaborn, Transformers, YOLO, Scikit-learn, SpaCy, NLTK, FastAPI, LangChain

Soft Skills: Leadership, Teamwork, Communication, Troubleshooting, Attention to Detail

PROFESSIONAL EXPERIENCE

QwertyZen, Kolkata: Python Application Developer

Feb 2026 – Present

- Developed a real-time cross-platform MIDI transcription app in Python, reducing note capture latency to 50ms.
- Implemented pitch detection pipeline using YIN algorithm, achieving 20 readings/sec with 99% monophonic accuracy.

Transcon IT Solutions Pvt Ltd, Kolkata: Multimodal Deep Learning Intern

Oct 2025 – Dec 2025

- Engineered a multimodal deep learning pipeline that converted 10,000+ sign images into synchronized audio outputs.
- Streamlined CV and speech synthesis models, enabling real-time gesture-to-text translation for 7,000+ images.

PROJECTS

Neuromatch - Neuromorphic Computing-Based Patient Retrieval

[GitHub Repo](#)

- Modeled cross-modal brain MRI similarity using a Siamese CNN with 128-d embeddings.
- Generated patient retrieval results achieving 0.71 mAP and 17% gain over ResNet-18 baseline.
- Used GPU-accelerated rank-order spike encoding processing 11,400 samples/sec across 370 patients.

Omguard – ML Powered Drone-Based Flood Prediction, Detection & Relief System

[GitHub Repo](#)

- Architected YOLO and DeepLabV3+ pipelines enabling drone-based flood detection and victim localization.
- Analyzed 10,000+ aerial frames achieving 92% detection accuracy during simulated disaster response scenarios.
- Integrated Ethereum blockchain facilitating transparent relief coordination across 50+ secure transactions per deployment.

Entropy-Based Sentence Boundary Detection

[GitHub Repo](#)

- Designed entropy-guided NLP framework leveraging knowledge graphs for context-aware sentence segmentation.
- Benchmarked performance across 5+ corpora, surpassing traditional tokenizers on precision metrics.
- Delivered Streamlit application enabling real-time visualization and parameter tuning for rapid experimentation.

RESEARCH WORK

AI-driven Monitoring System for Detecting People Using Mobile Phones in Restricted Zones

[Paper Link](#)

- Designed deep learning pipeline using YOLOv8x and YOLOv8-ResNetv2 for mobile phone detection.
- Applied image preprocessing, bounding-box refinement, and DBB spatial proximity algorithm for behavior inference.
- Evaluated on 1,200+ surveillance frames achieving 84% precision, 72% recall, and competitive mAP scores.

EDUCATION

Heritage Institute of Technology

CGPA: 8.45

B.Tech in CSE (AI & ML), Expected 2027, Kolkata

The Aryans School

ICSE (10th): 97.6% ISC (12th): 92.75%

2009–2023, Kolkata

ACHIEVEMENTS

- Winner, Hack4Bengal 3.0 Virtual Hack 2025 (200+ teams)
- 1st Position, Exabyte Encode 2025, St. Xavier's College, Kolkata
- 2nd Runner-up, Hack Synthesis 2024, UEM Kolkata
- 3rd Runner-up, Innovathon 2024, NSHM Kolkata
- 3rd Runner-up, Hack Fusion 2024, NIT Jamshedpur